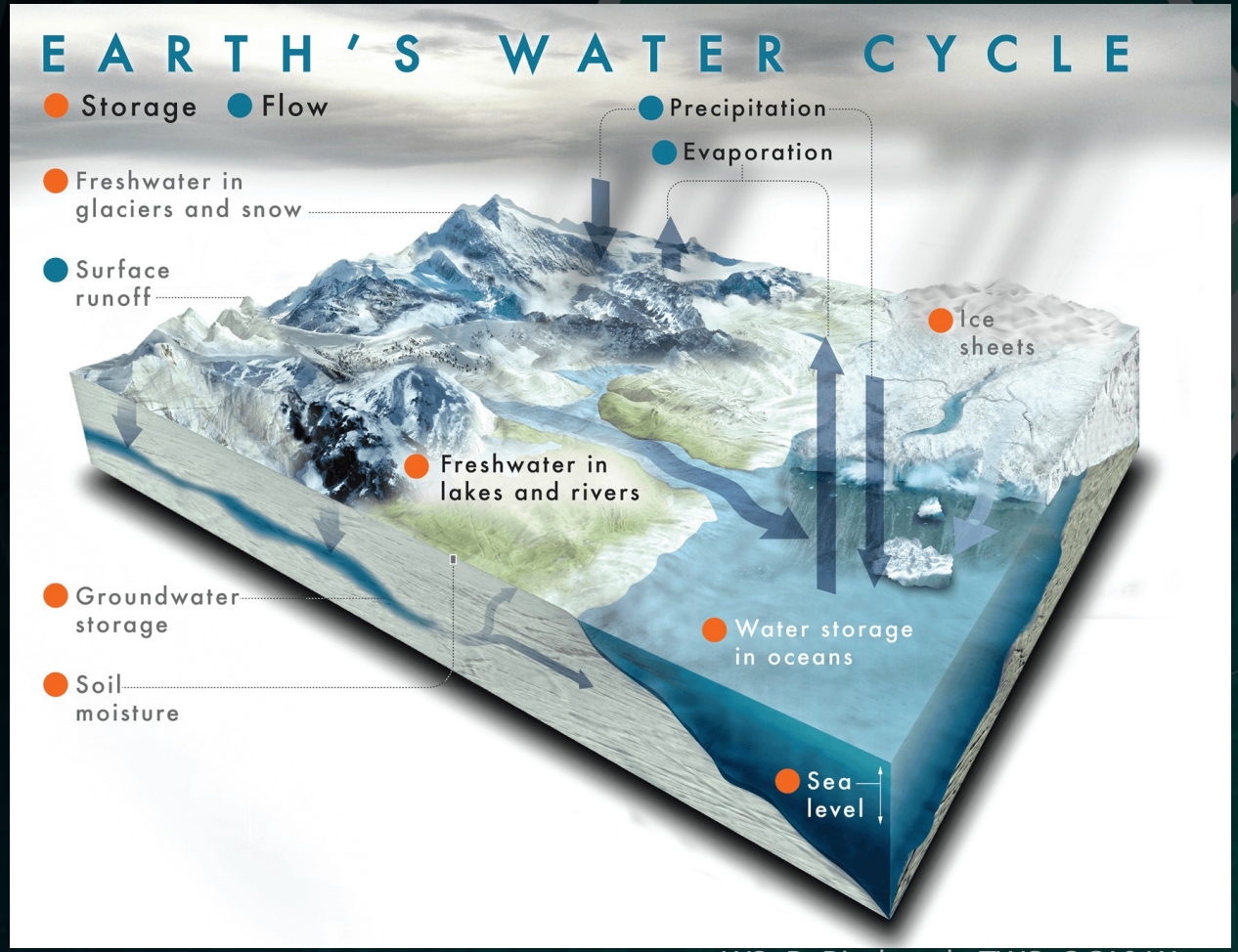
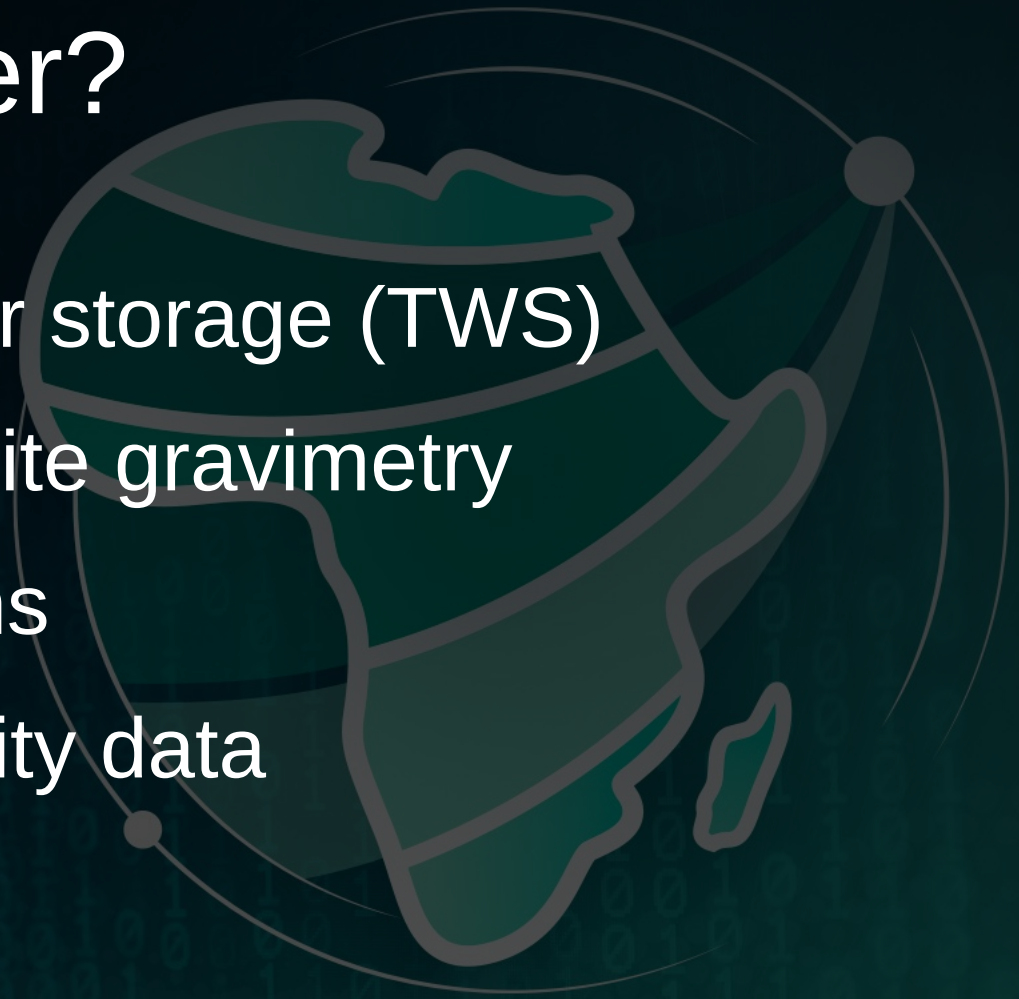
W3: Terrestrial Water Storage

OC10: Water budgets from Space, R. Rietbroek



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What will we cover?



- Concept of terrestrial water storage (TWS)
- Observing TWS with satellite gravimetry
- Current and future missions
- Working with satellite gravity data

Concept of terrestrial water storage (TWS)



Terrestrial Water Storage is an essential climate variable (ECV)

- TWS is an official GCOS ECV since 2022
- ECV's:
  - 55 defined by GCOS (Global Climate Observing System)
  - variables that critically contributes to the characterization of Earth's climate
  - Strong incentive for strategic Monitoring/modelling (e.g. helps in directing mission design)



Terrestrial Water Storage (TWS)

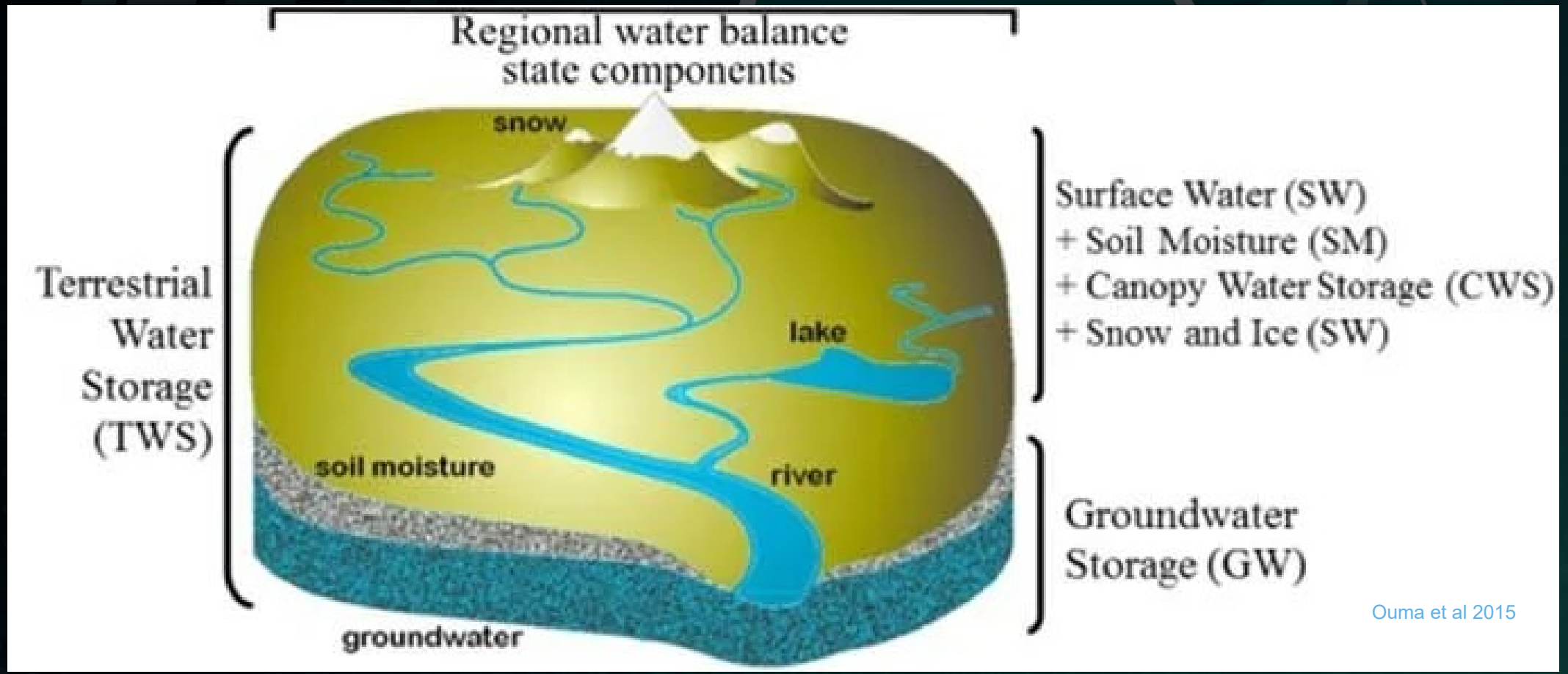
TWS is the total amount of water stored in all continental storage compartments (ice caps, glaciers, snow cover, soil moisture, groundwater, surface water bodies, water in biomass). The change of TWS over time balances the budget of the water fluxes precipitation, evapotranspiration and runoff, i.e., it closes the continental water balance.

Terrestrial Water Storage is an essential climate variable (ECV)

TWS is the total amount of water stored in all continental storage compartments (ice caps, glaciers, snow cover, soil moisture, groundwater, surface water bodies, water in biomass). The change of TWS over time balances the budget of the water fluxes precipitation, evapotranspiration and runoff, i.e., it closes the continental water balance. TWS changes thus represent the net effect of climate variability and climate change as well as of anthropogenic impacts on the water cycle.

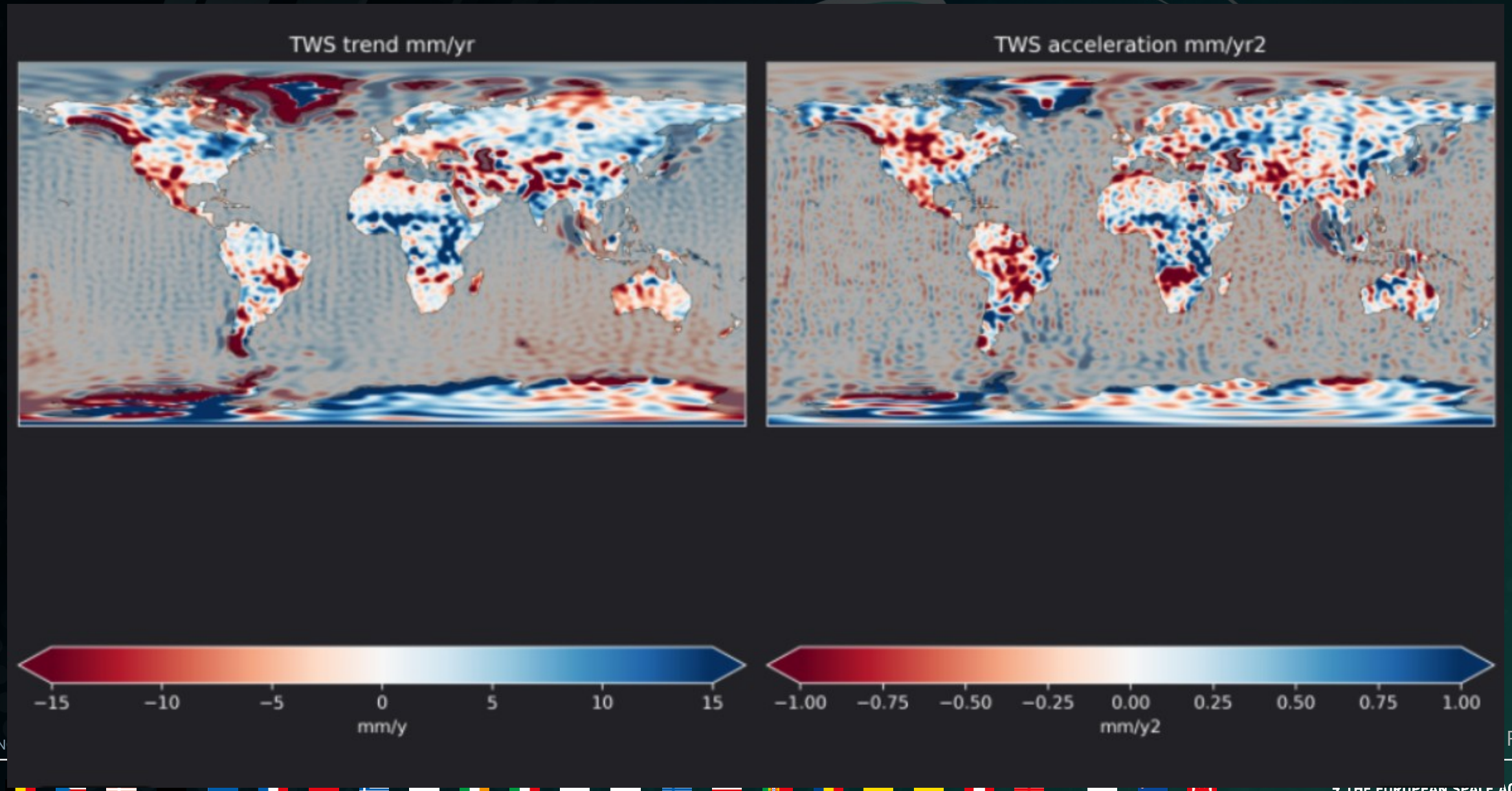
- TWS is an aggregate measure of continental water storage (liquid/solid)
- It's time derivate is a **FLUX** (e.g. in m3/s)
- Often expressed in m (layer of equivalent water height)
- Typical variations: 0-40mm of water height change
- Time scales: days - years

Tws is an aggregate variable

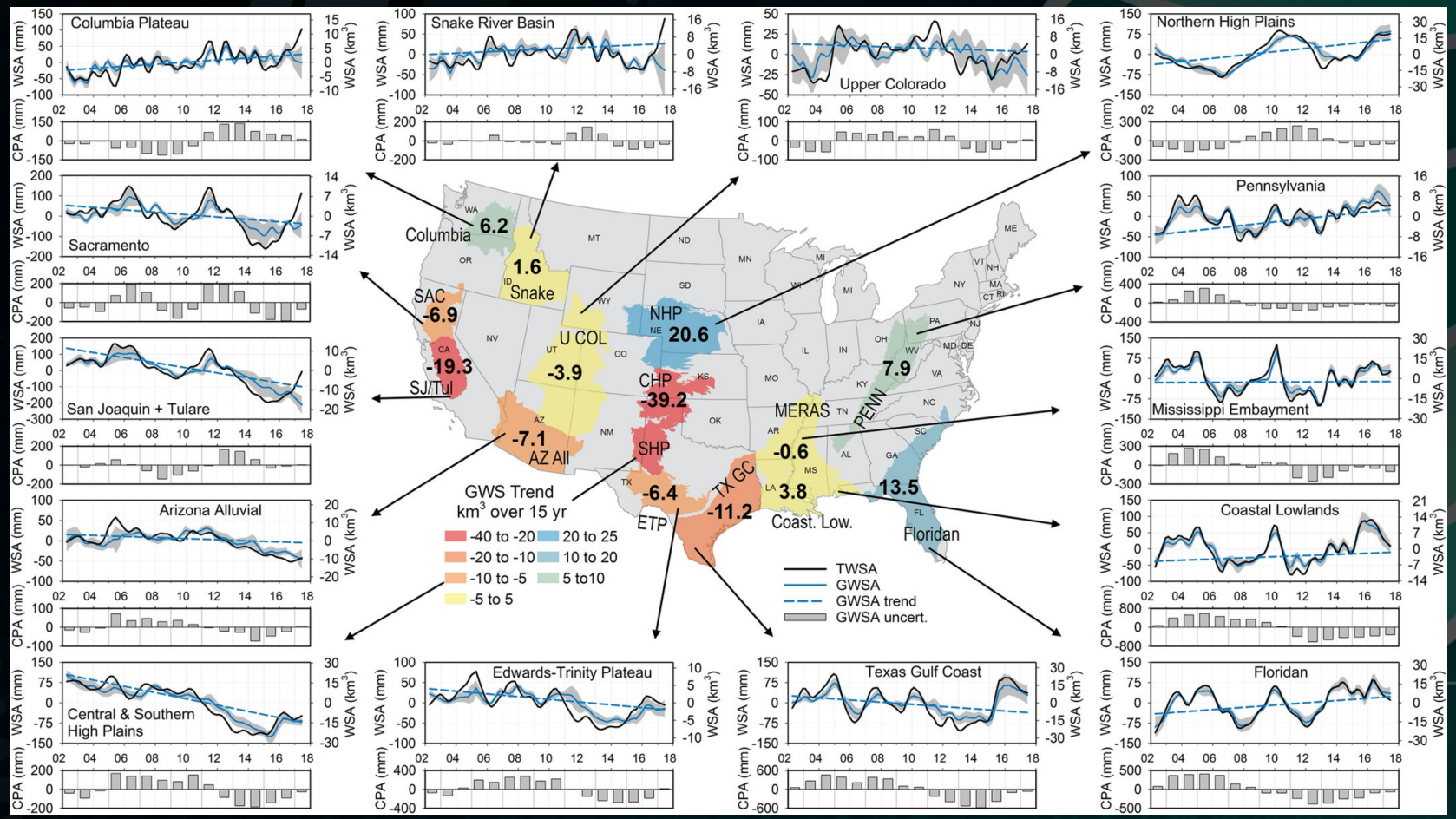


Ouma et al 2015

Can you spot TWS trend hotspots?



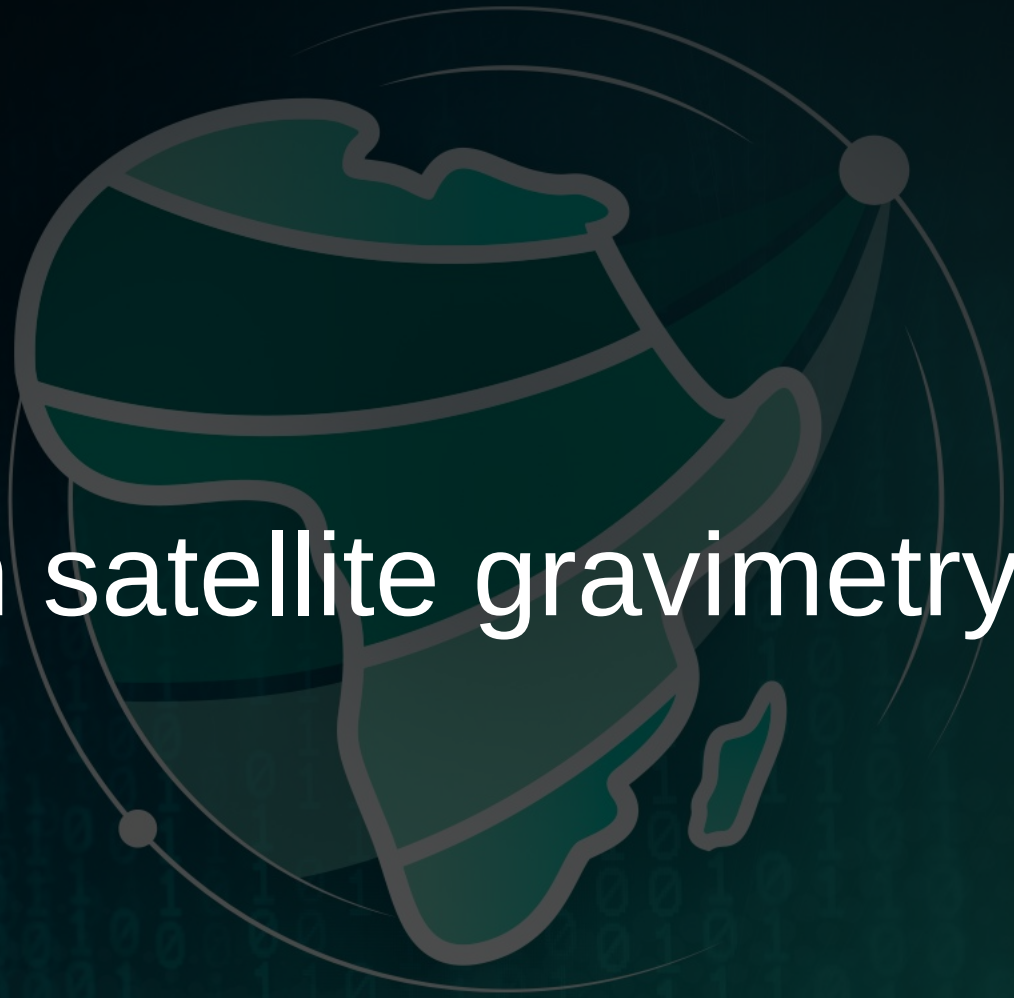
Example of using TWS: US Groundwater changes from GRACE/GRACE-FO



From Rateb et al. 2020, Water resources research. WSA: water storage anomaly, CPA:cumulative precipitation anomaly

- Aquifer mass changes can be observed from gravity changes
- Net US decrease 90 km^3 over 15 years, but signs vary per aquifer
- Groundwater peaks (Blue) are damped relative to terrestrial water storage (black), why?
- **Retrieving groundwater from TWS requires stripping away other components (soild moisture , surface water, snow,..)**
- GRACE/GRACE-FO are gravity missions. How can they be sensitive for TWS?

Observing TWS with satellite gravimetry



How does gravity relate to TWS? → quick intro to geodesy

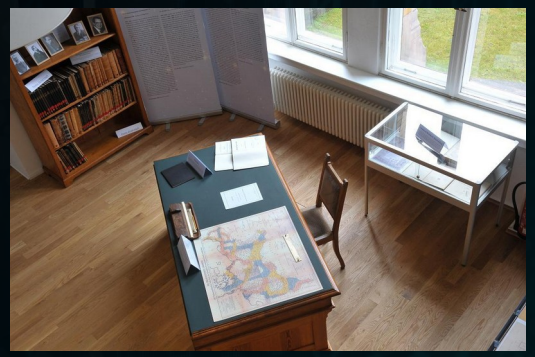


- We're not talking about the Hollywood "gravity"

Classical definition of Geodesy



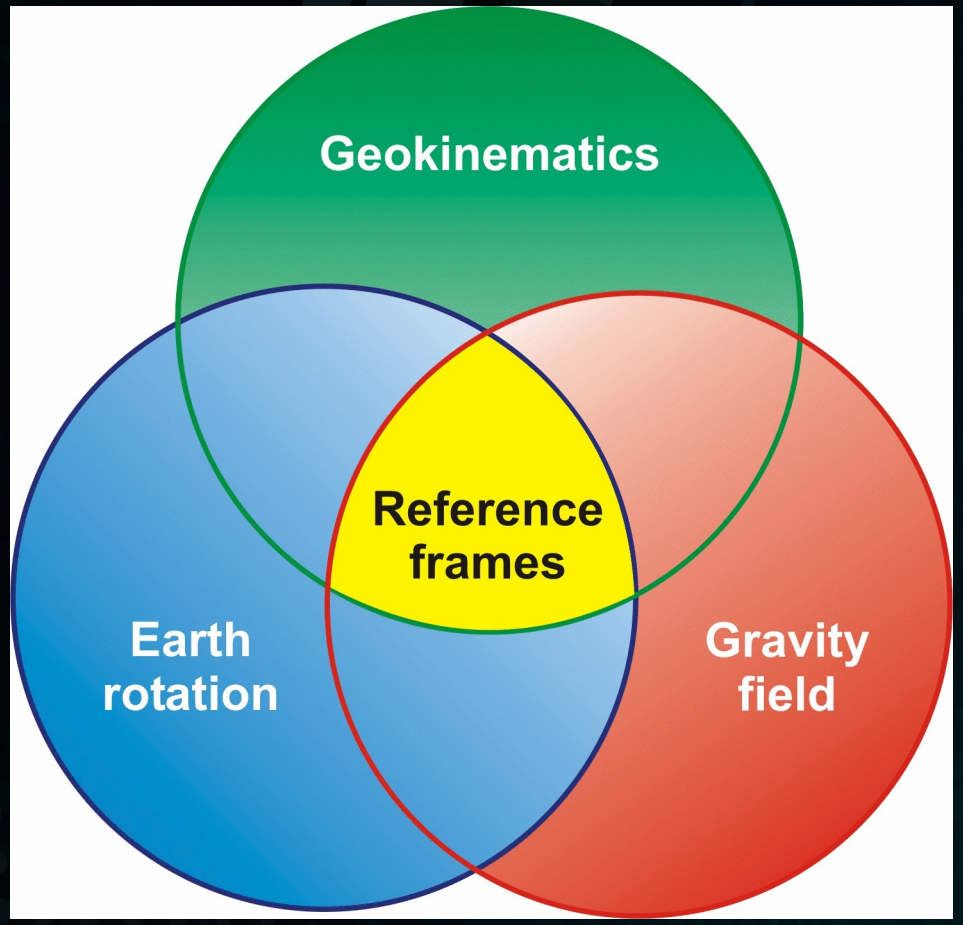
Friedrich Robert Helmert



Helmert's desk (GFZ, Potsdam)

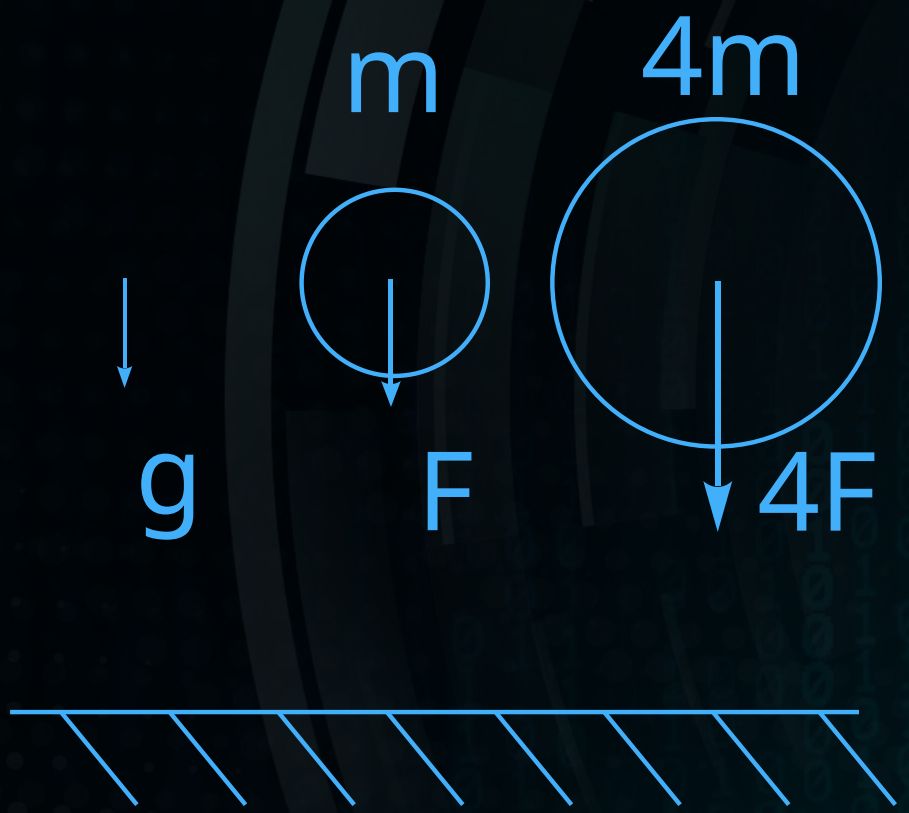
- Friedrich Robert Helmert (1880): Geodesy is the science concerned with the study of the shape and size of the earth in the geometric sense as well as with the form of the equipotential surfaces of the gravity potential
- Still relevant and well ahead of his time

The three pillars of Geodesy



- **Geokinematics**: study the shape of the Earth and its deformations over time)
- **Earth rotation**: position and rate of the rotation axis and angular momentum axis w.r.t. to stars and crust
- **Gravity**: the (possibly time-variable) gravitational field of non-homogenous objects
- **Reference frames**: Definition and realization of Celestial and or Earth-fixed reference systems/frames

Force versus Acceleration



- Gravitational force for objects in free fall: $F = mg$
- To change the magnitude or direction of an object's velocity, a force is needed
- Force equals (gravitational) acceleration times mass
- Gravitational acceleration is independent of the mass of the (falling) object
  - But depends on mass of attracting body
- It's more convenient to describe the gravity field of an attracting body in terms of acceleration!
- How to determine gravity **g**?

What influences gravity?

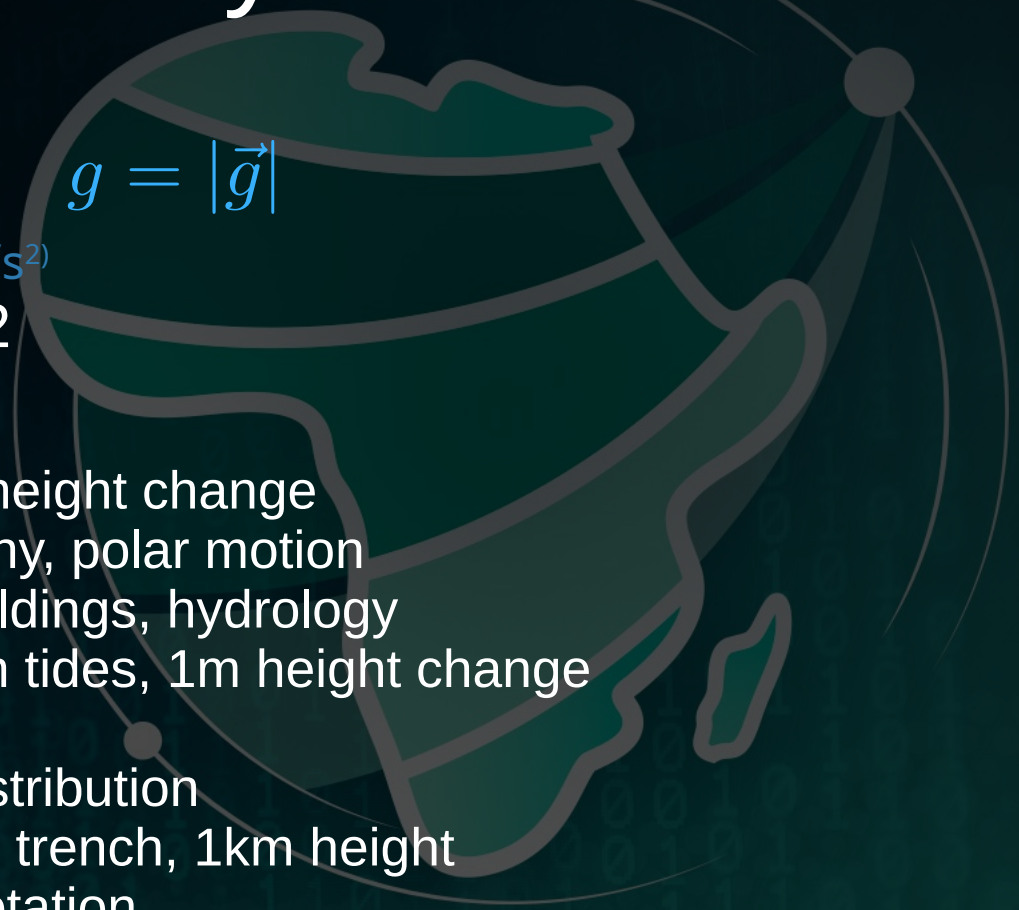
- Magnitude of gravity

$$g = |\vec{g}|$$

Gal mGal μGal (1 Gal =1cm/s²)

9.807246731 … m/s²

- Relativity, 1mm height change
- Ocean topography, polar motion
- Nearby large buildings, hydrology
- Earth and Ocean tides, 1m height change
- Large reservoirs
- Internal mass distribution
- Mountain, ocean trench, 1km height
- Flattening and rotation

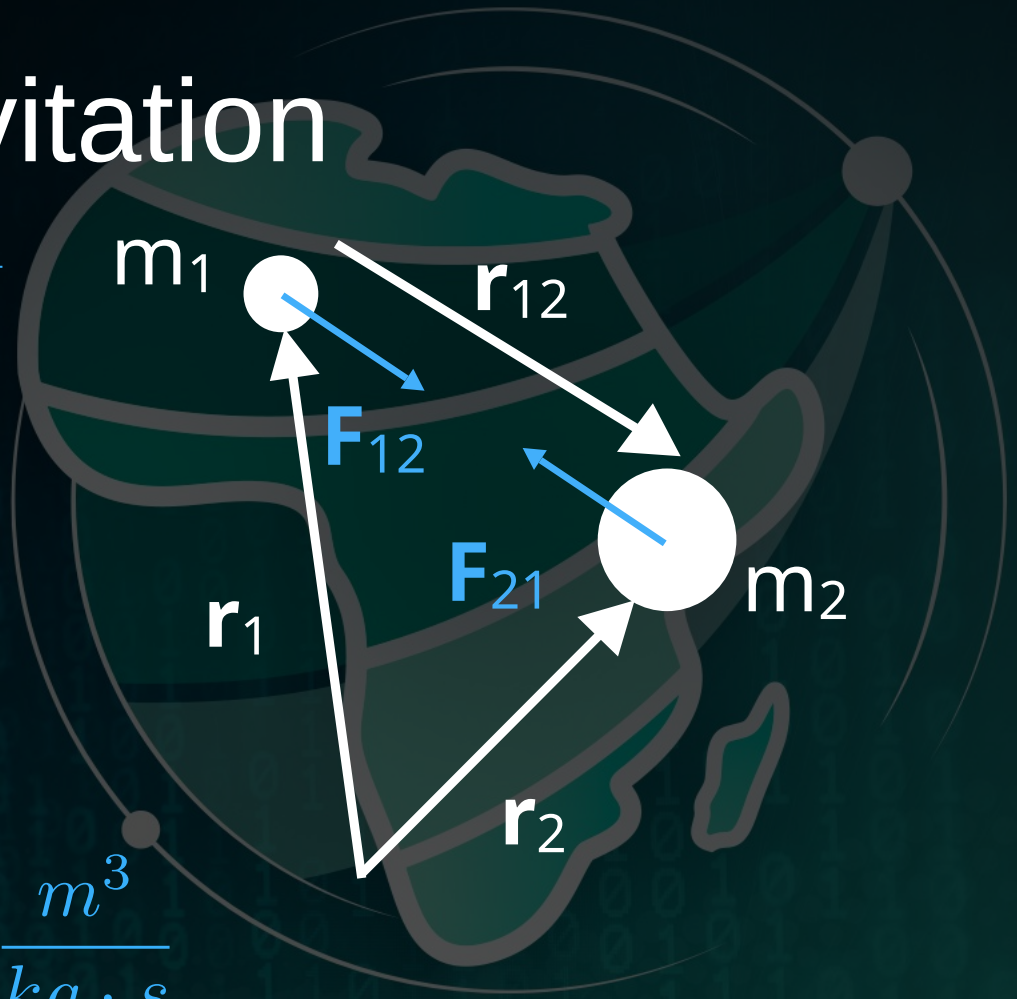


Newton's law of gravitation

- Gravitational force → $\mathbf{F}_{12} = \frac{G m_1 m_2 \mathbf{r}_{12}}{|\mathbf{r}_{12}|^3}$
  proportional to both masses
  Inversely-proportional to square of distance
- Action = Reaction →
- Gravitational constant →

$$\mathbf{F}_{12} = -\mathbf{F}_{21}$$

$$G = 6.67384 \times 10^{-11} \frac{m^3}{kg \cdot s}$$

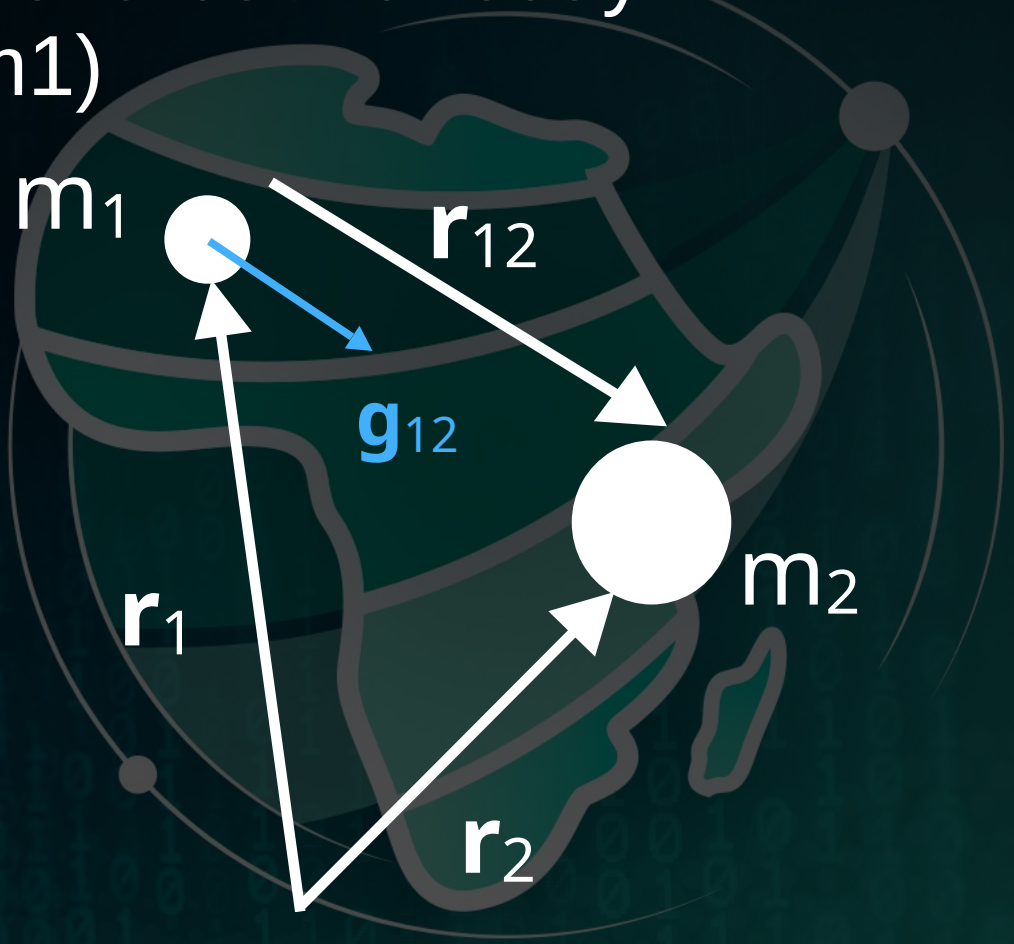


Describe the gravitational field of a central body in terms of acceleration (m2 >> m1)

- Acceleration for a point mass:

$$\mathbf{g}_{12} = \frac{G m_2 \mathbf{r}_{12}}{|\mathbf{r}_{12}|^3}$$

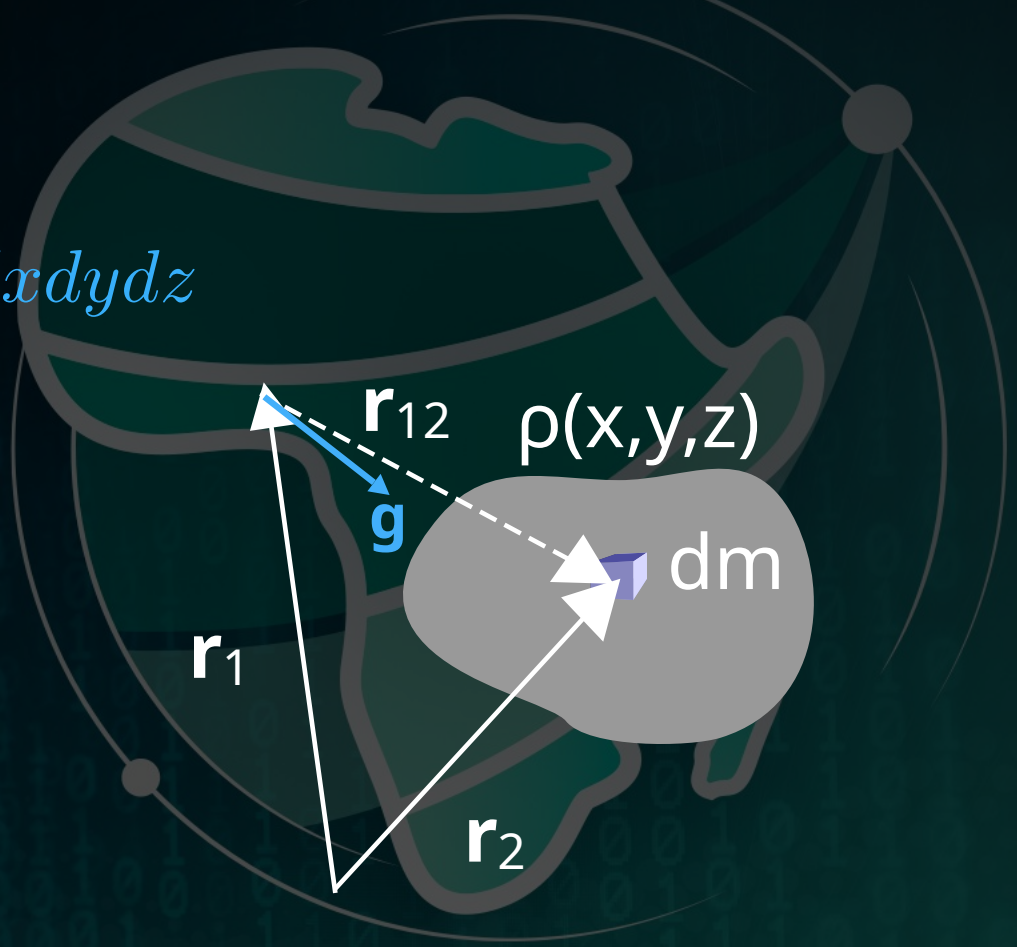
- More complex bodies: apply superposition principle
  - Addition of multiple point masses
  - Introduction of a density function



Integrate density to obtain the gravitational field

$$\mathbf{g} = \iiint_{\Omega} \frac{G\mathbf{r}_{12}}{|\mathbf{r}_{12}|^3} \rho(x,y,z) dx dy dz$$

- Addition of multiple point masses
- Introduction of a density function (uh oh, do we even know this?)
- g points along local **plumbline**
- Is there a better choice for the reference frame origin?

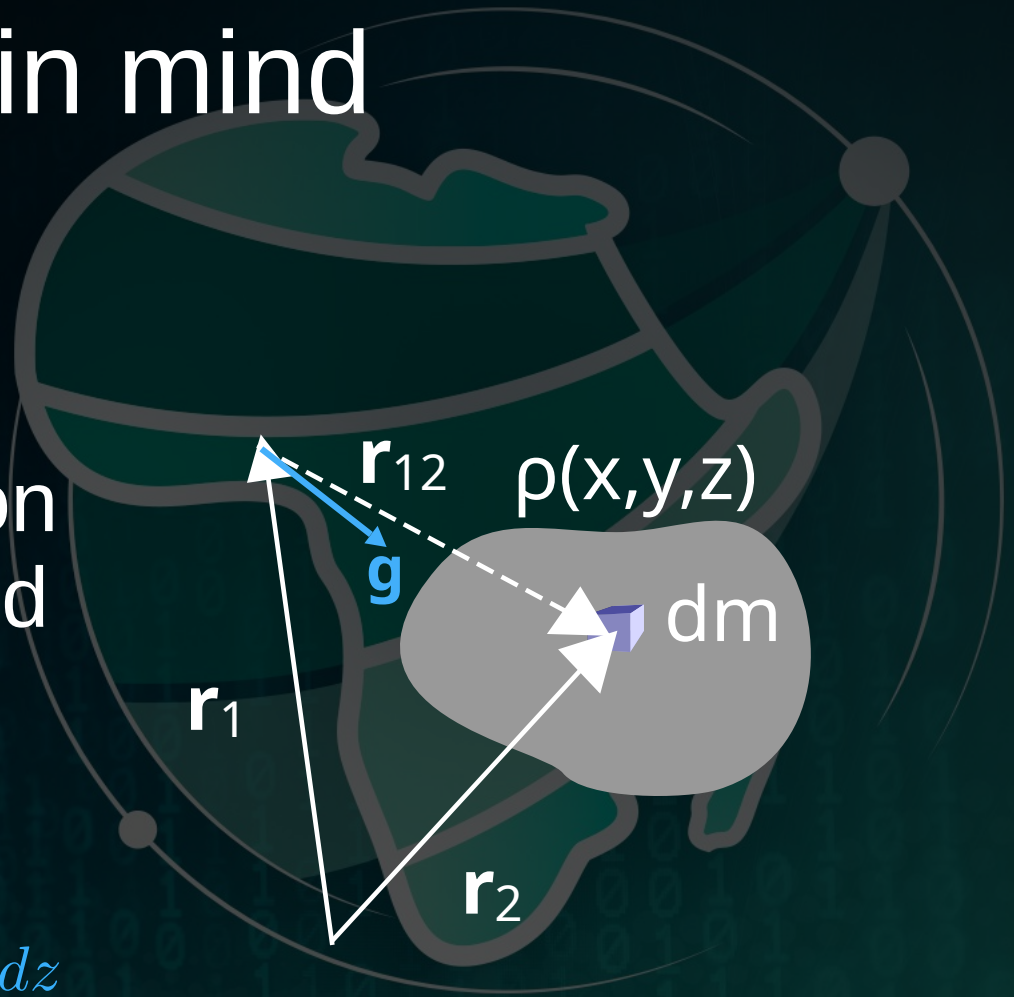


Key idea to keep in mind

$$\mathbf{g} = \iiint_{\Omega} \frac{G\mathbf{r}_{12}}{|\mathbf{r}_{12}|^3} \rho(x,y,z) dx dy dz$$

The Earth's mass distribution changes over time (TWS!) and thus create a time variable gravity field:

$$\mathbf{g}(x,y,z,t) = \iiint_{\Omega} \frac{G\mathbf{r}_{12}}{|\mathbf{r}_{12}|^3} \rho(x,y,z,t) dx dy dz$$



Properties of g?

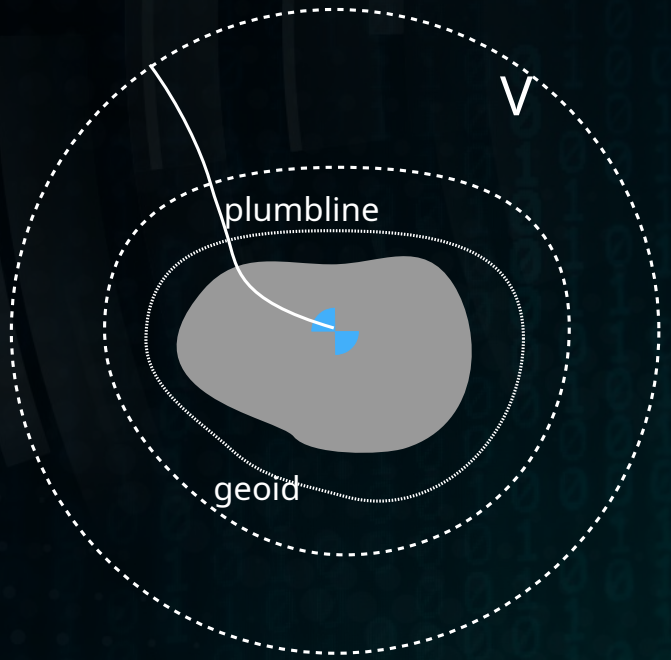


- Does g point to the center of mass?
  - only in special cases
  - from infinity
- How quickly does g vanish going to infinity?
  - with the second power of distance from the center of mass
- Does g generate a *conservative force*?
  - Yes, it must therefore generate a potential field
  - gravity follows plumblines and is orthogonal to equipotential surfaces
  - the geoid is special (coincides with an ocean surface at rest)

Potential versus Gravity

Potential V
- Scalar field
- unit m^2/s^2
- not directly observable
- Mathematical aid

Gravity **g**
- Vector field
- unit m/s^2
- Observable
- Gradient of the Potential



$$\nabla V = \mathbf{g}$$

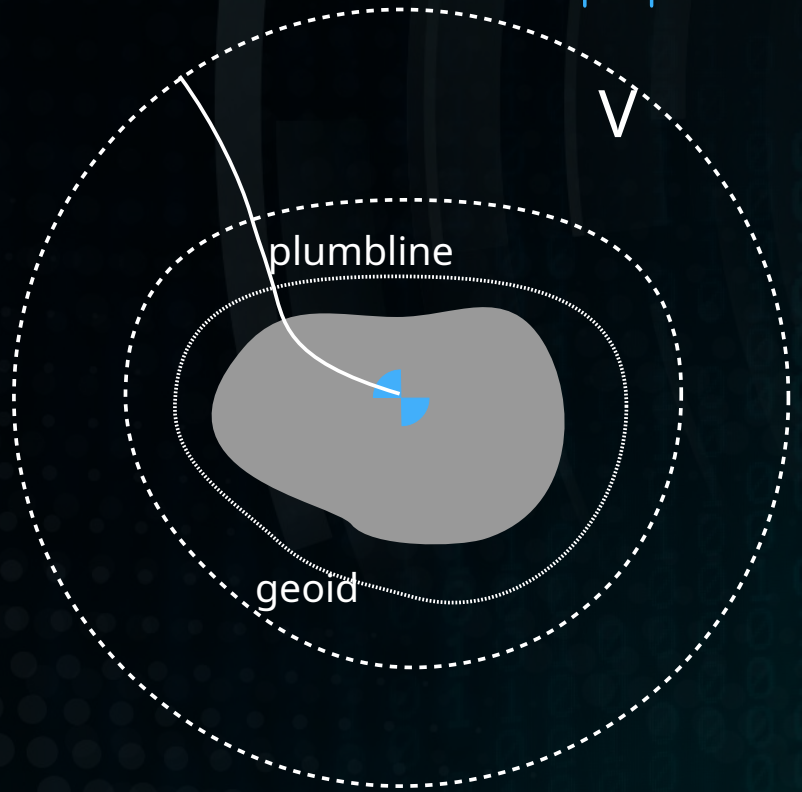


Potential versus gravity from infinite distance

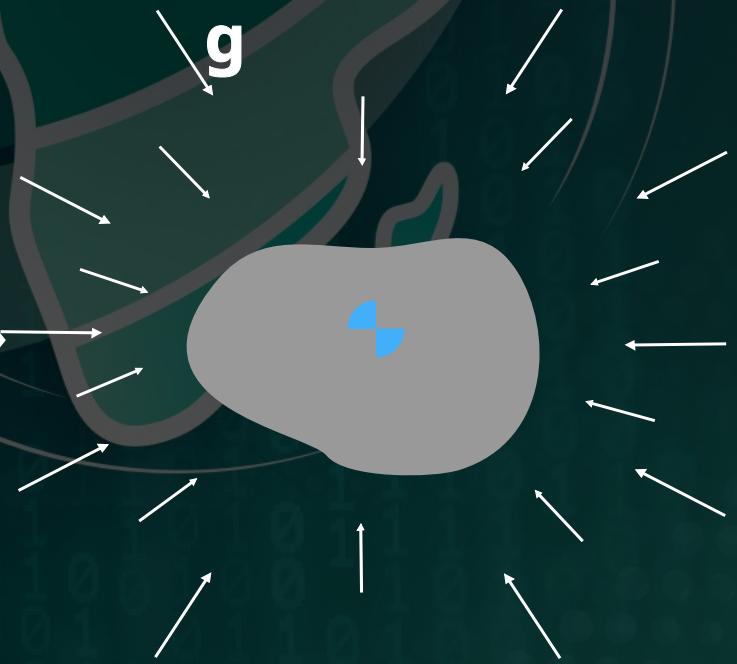
No matter how complex the body is: from a distance it looks like a point mass

$$\lim_{r\to\infty} V = \frac{GM}{|\mathbf{r}|}$$

$$\lim_{r\to\infty} \mathbf{g} = -\frac{GM\mathbf{r}}{|\mathbf{r}|^3}$$



$$\nabla V = \mathbf{g}$$

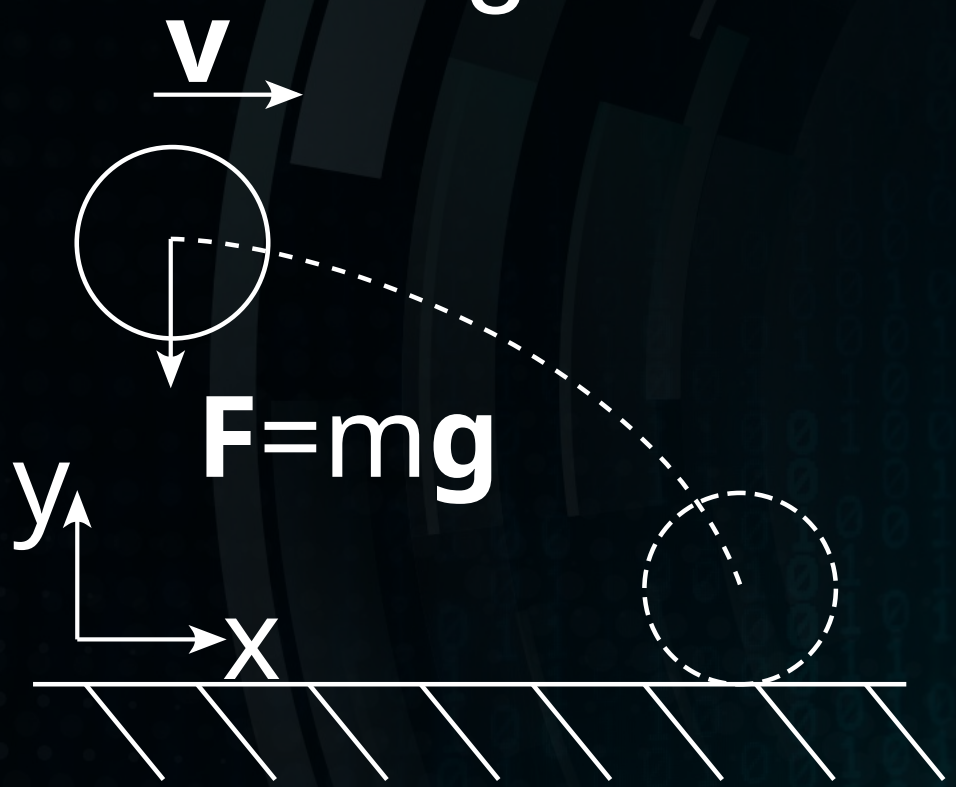


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How does gravity affect orbits?



Throwing a stone



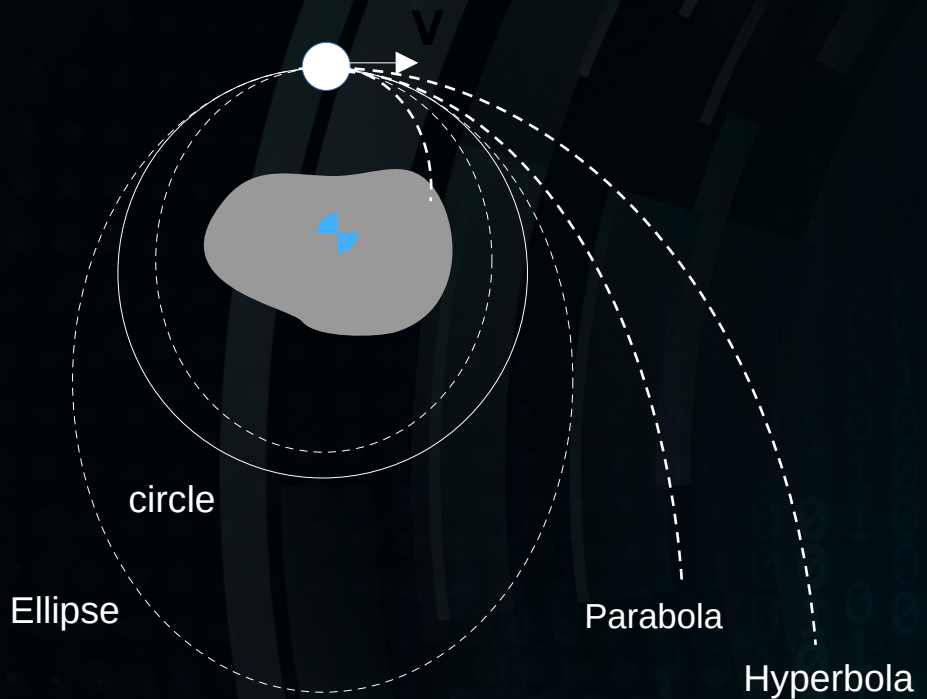
- What is the shape of the path?
  - Parabola is a good approximation

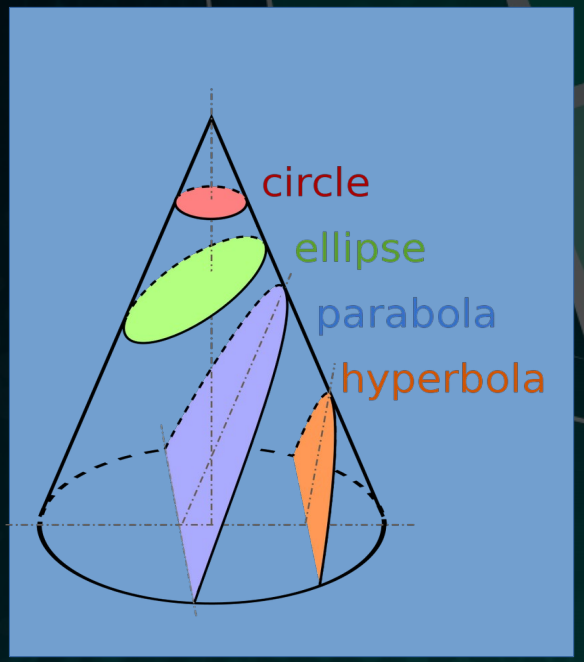
$$x(t) = x_0 + Vt$$

$$y(t) = y_0 - \frac{1}{2}gt^2$$

- but g is not constant (neither magnitude or direction)
- What happens if we increase horizontal velocity, V?

Orbital paths





- With increasing V
  - Orbit
  - Escape Orbit
- Earth as a point mass
  - Kepler orbit
  - Conical sections

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How do we know? Painstaking work from Tycho Brahe and Johannes Kepler



Tycho Ottesen Brahe

Brahe wearing the Order of the Elephant

Born	14 December 1546 Knutstorp Castle, Scania, Denmark, Denmark–Norway (now Sweden)
Died	24 October 1601 (aged 54) Prague, Habsburg Bohemia, Holy Roman Empire (now Czech Republic)
Nationality	Danish
Alma mater	University of Copenhagen Leipzig University University of Rostock
Occupation	Nobleman, astronomer, writer
Known for	Tychonic system *Rudolphine Tables*
Spouse(s)	Kirsten Barbara Jørgensdatter
Children	8
Parent(s)	Otte Brahe Beate Clausdatter Bille

Signature



Johannes Kepler

Portrait of Kepler by an unknown artist, 1610

Born	December 27, 1571 Free Imperial City of Weil der Stadt, Holy Roman Empire
Died	November 15, 1630 (aged 58) Regensburg, Electorate of Bavaria, Holy Roman Empire
Residence	Württemberg; Styria; Bohemia; Upper Austria
Nationality	German
Alma mater	Tübinger Stift, University of Tübingen
Known for	Kepler's laws of planetary motion Kepler conjecture *Rudolphine Tables*
Scientific career	
Fields	Astronomy, astrology, mathematics and natural philosophy
Institutions	University of Linz
Doctoral advisor	Michael Maestlin
Influences	Nicolaus Copernicus Tycho Brahe
Influenced	Sir Isaac Newton

Signature

wikipedia

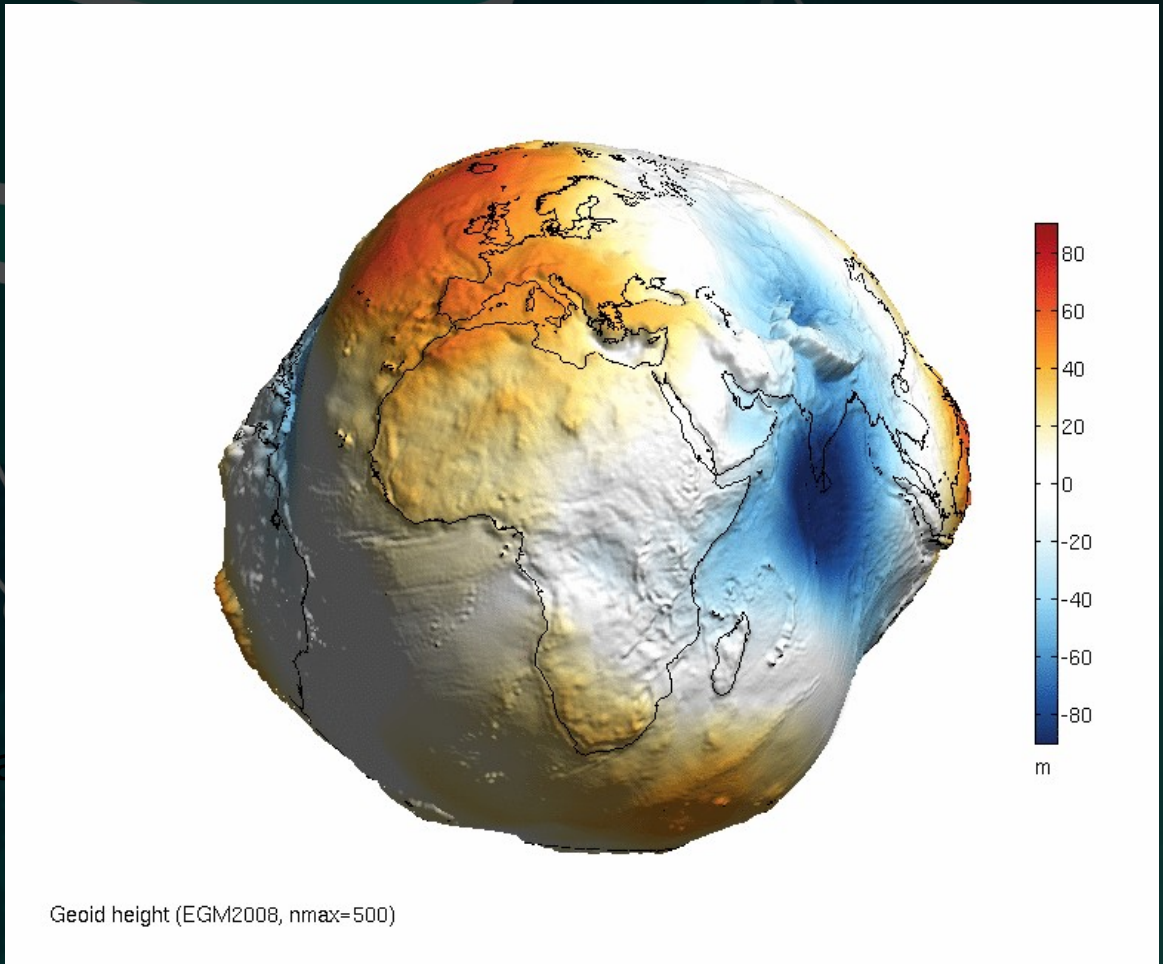
- Tycho Brahe → compiled a comprehensive database of astronomical observations
- Johannes Kepler → explained these astronomical observations with his 3 laws of planetary motion
- Kepler was the assistant and follow up of Brahe during his time in Prague

Intermediate Lessons learned

- Geodesy is build around three pillars: Shape,earth rotation and gravity
- We can describe the gravitational field in terms of acceleration
- Or better: in terms of gravitational potential from which we can derive the vectorial gravity anywhere
- The potential field can visualized by equipotential surfaces wrapped around the central body
- Orbits around point masses (e.g. Earth around the Sun) are described by Kepler orbits (conical sections)

The Earth as a potato (aka Geoid)

- Geoid (surface of the ocean in rest sea without motion)
- Natural choice as a reference surface (where is 0m height?)
- max -100 meter difference from Ellipsoid
- Equipotential surface: Paths along the geoid do not involve work
- Visible in the geoid: Tectonics, mantle convection, bathymetry, topography

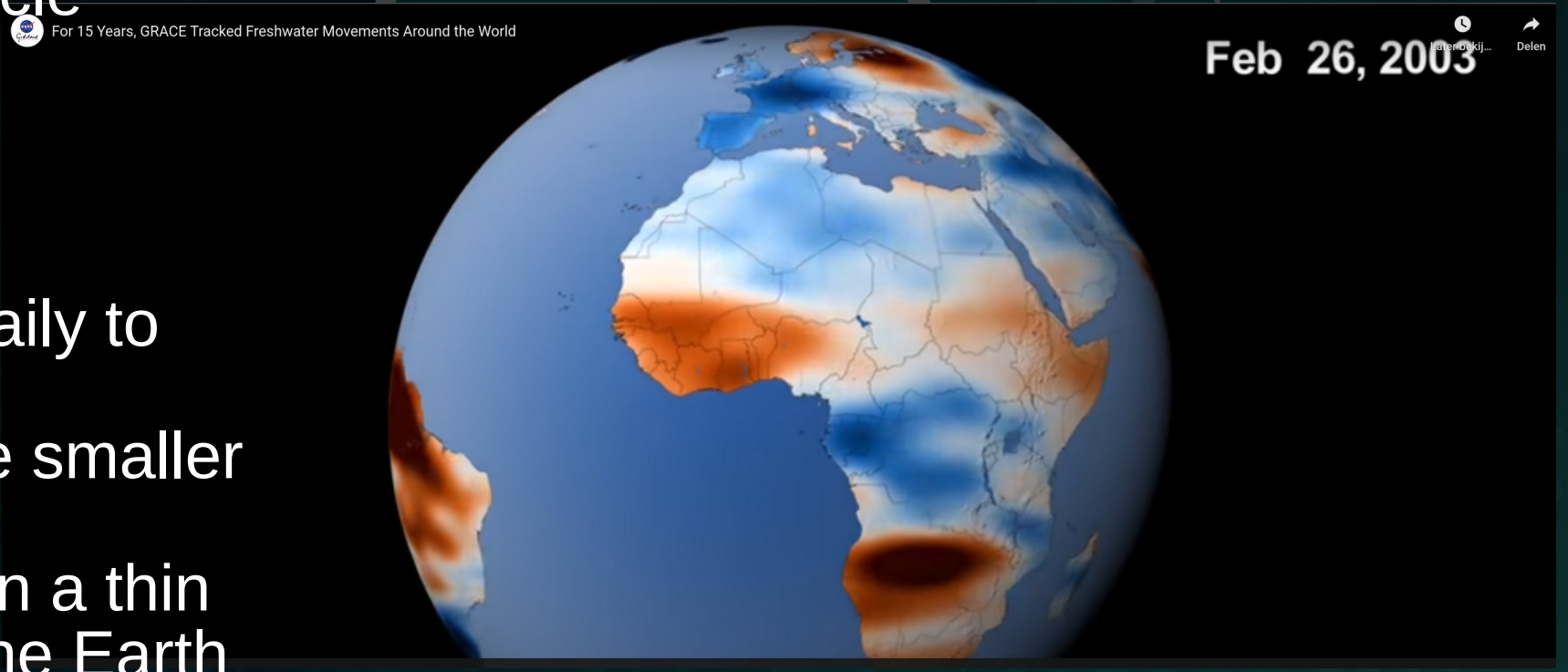


Geoid height (EGM2008, nmax=500)

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But there is a time variable component!

- Tides
- Hydrological water cycle
- Atmosphere
- Ocean
- Solid Earth (e.g. Earthquakes,..)
- Highly variable sub-daily to inter-annual
- 3 orders of magnitude smaller than the static field
- Most changes occur in a thin layer at the surface of the Earth (~15km thick)



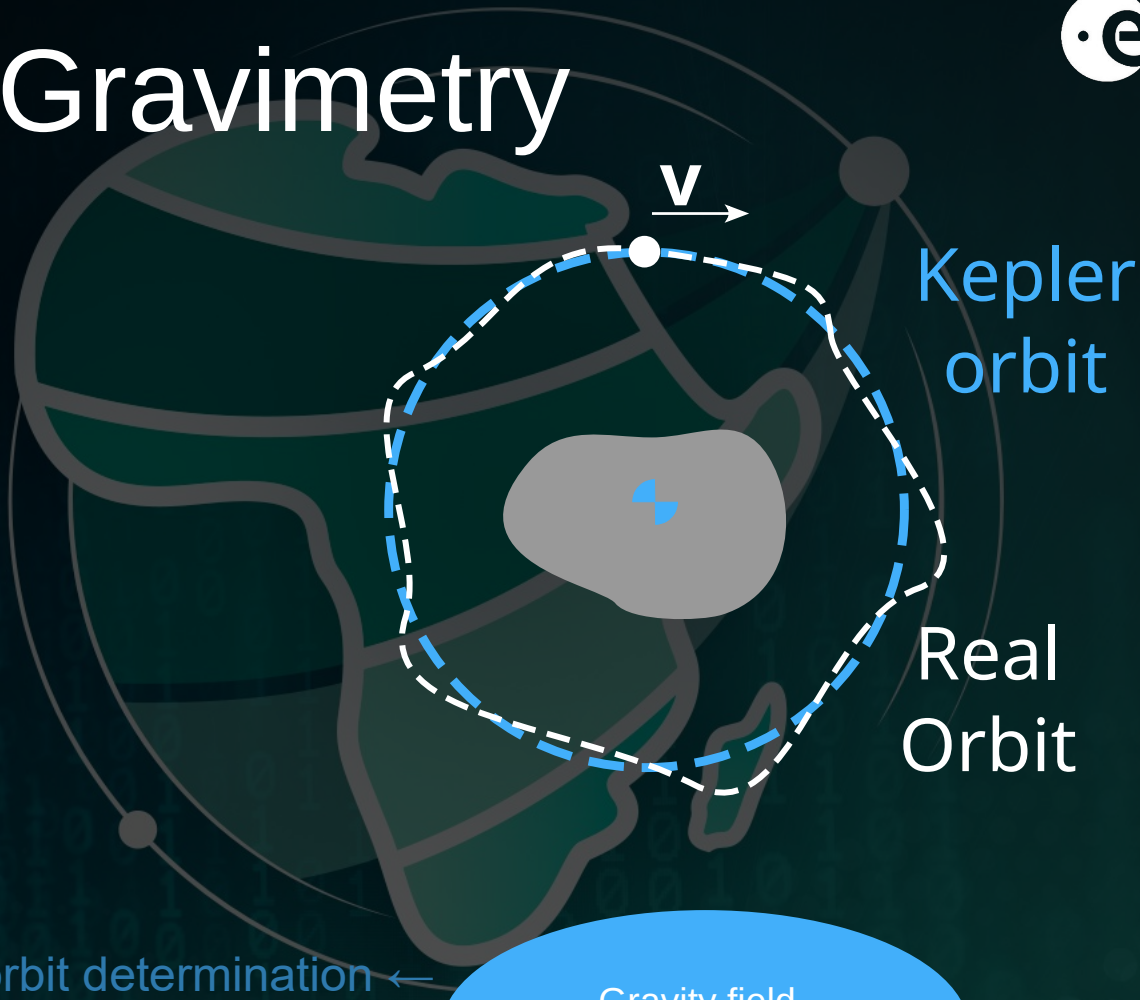
video link

Video opens up more questions

- How can satellite gravimetry measure water height changes?
  - after all, there is an infinite number of density distributions which generate the same external gravity field
- How does one separate all those different contributions (soil moisture, surface and groundwater, atmosphere)?
- Why does the gravity field from GRACE look so smoothed?

Principle of Satellite Gravimetry

- Anomalies from a perfect Kepler orbit contain gravity field information
- Anomalies decreases for higher orbital altitudes
- Largest gravity anomaly?
  - The Earth's flattening



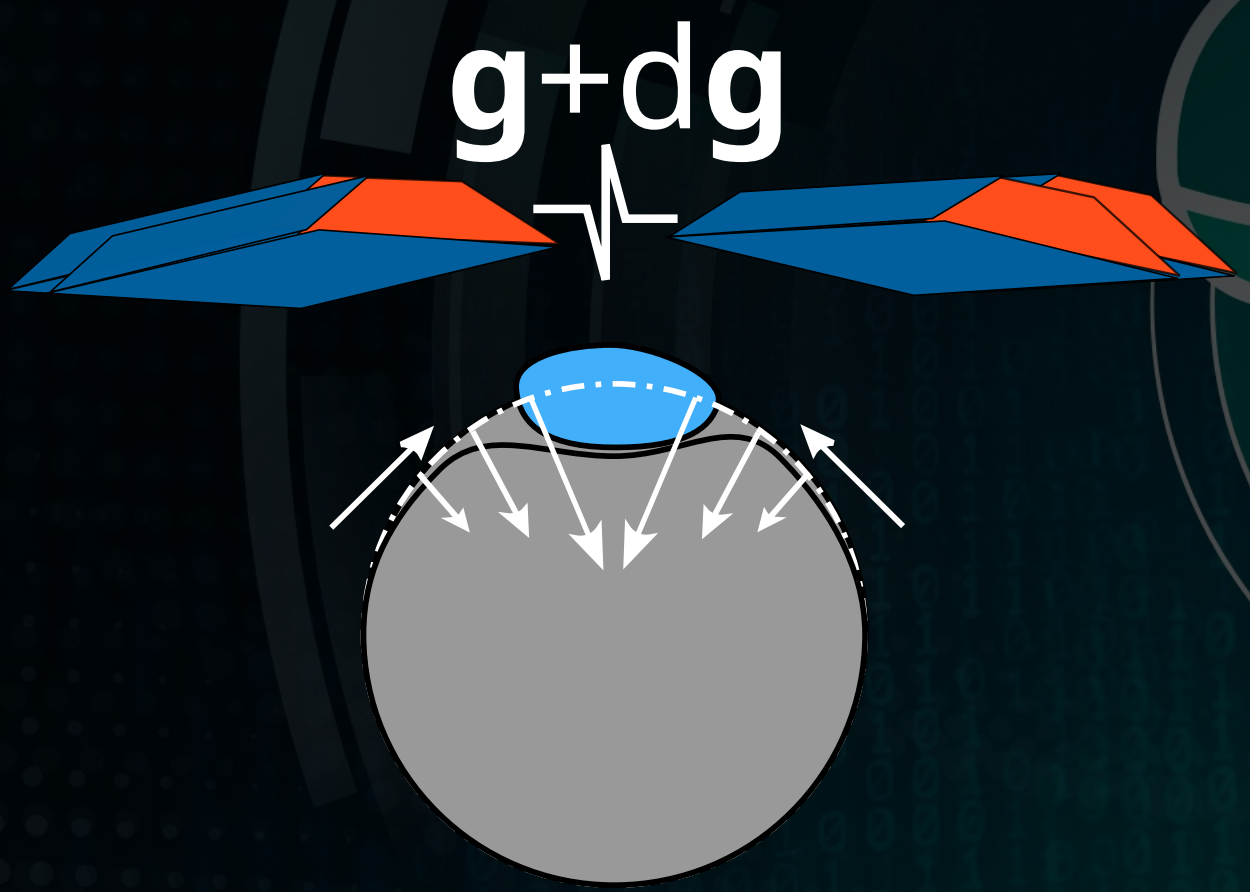
Satellite orbit (position, velocity)

← Precise orbit determination ←

→ Gravity field estimation →

Gravity field (potential, gravity)

Surface load (e.g. TWS) and satellite gravimetry



- Surface load causes
  - A **direct** change in gravity
  - an **indirect** response due to deformation of the solid Earth
- Terrestrial (total) Water Storage can be retrieved from satellite orbits when:
  - We assume mass change occurs in a thin layer
  - We have a dense coverage of accurate orbit measurements (e.g. from GNSS or range instruments)
  - We know the mechanical properties of the solid Earth

Current and future gravity missions



GRACE Mission (2002 - 2017)



- Launch March 2002, Plesetsk (Rockot, reused ICBM)
- End Nov/Dec 2017
- Orbit height: 500km (350 km during last phase)
- Satellite separation:~250km
- Orbital period: 94 min
- Inclination: 89 degs

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GRACE Mission cont'd



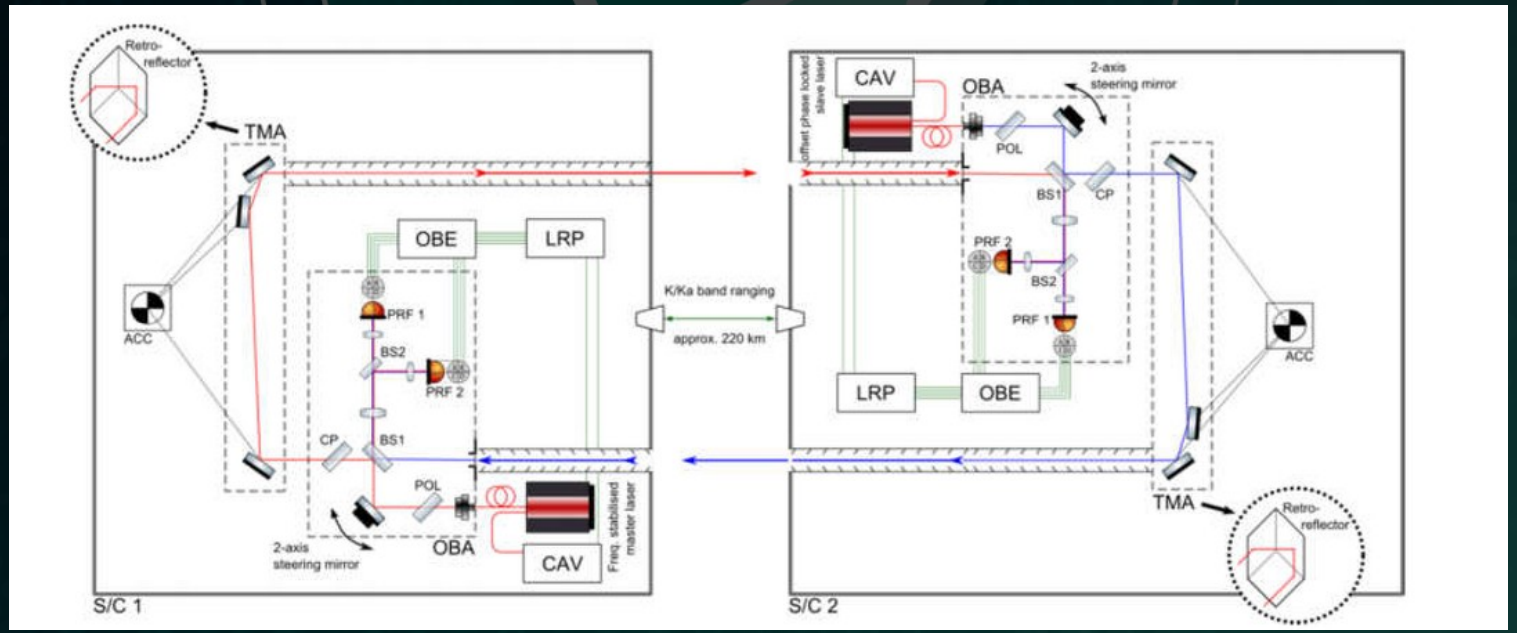
- Mission life time: 5 yrs!
- Satellite weight: 460kg+34kg fuel
- Power 160W
- Microwave ranging instrument (σ < 1 μm/s)
- Accelerometer (σ = 3e-10 m/s2)
- Global positioning System (σ = 2 - 3 cm)
- NASA: 97 million USD, DLR: 30 million USD

GRACE Follow on Mission

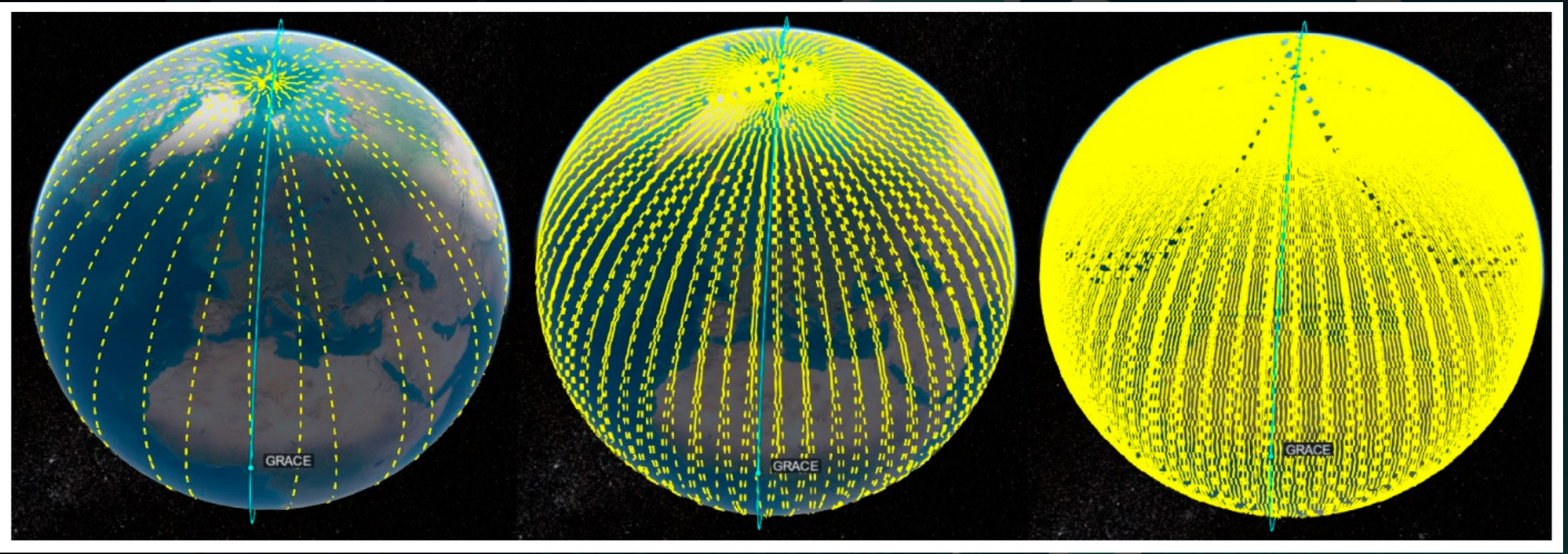




- Launched: May 22, 2018
- Piggy-back with Iridium satellites on Falcon launcher
- Demonstrator: Laser interferometer for intersatellite
- ranging (~10x more accurate)



Sampling/groundtracks



Fatolazadeh et al. 2022

- GRACE and GRACE-FO are in a polar orbit
- Spatial resolution is linked to averaging interval
- Short term hydrology poses challenges for satellite gravimetry in terms of spatiotemporal resolution

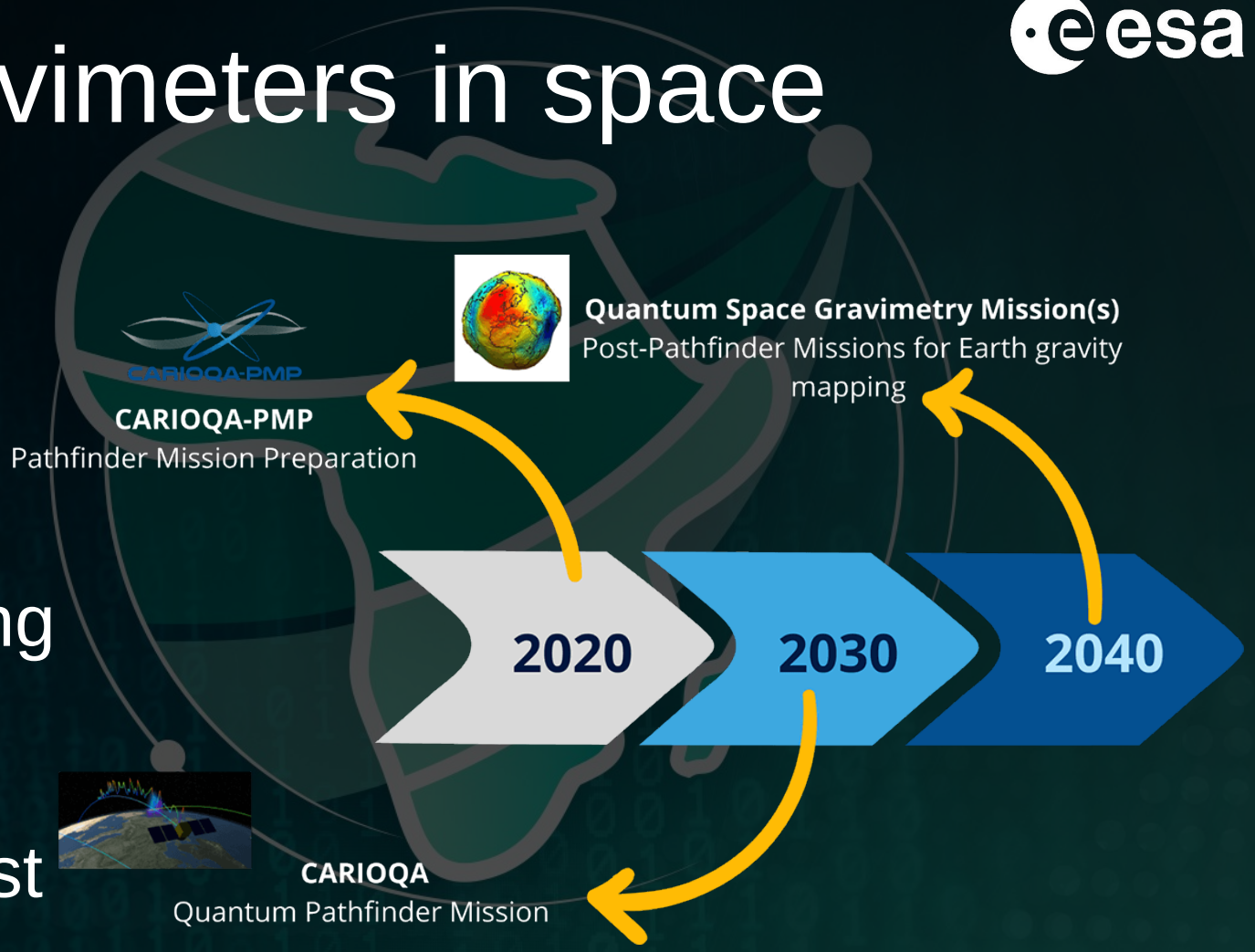
MAGIC (GRACE-C + NGGM)

- 2 upcoming satellite pairs (Collaborate with NASA)
  - GRACE-C (Continuity) (NASA-DLR) 2028
  - NGGM (ESA), 2032
- Mass-change And Geosciences International Constellation (MAGIC)
- Two simultaneous pairs at different inclination (‘Bender’ configuration)
- @ 150 km spatial resolution in 10 days
- @300km in 3 days
  - Relevance for shortlived events such as flooding



Quantum gravimeters in space

- Accelerometer noise is currently a bottleneck
- Quantum (Cold Atom Rubidium Interferometers) based accelerometers are being developed for future missions
- Launch Pathfinder mission before end of the decade to test potential

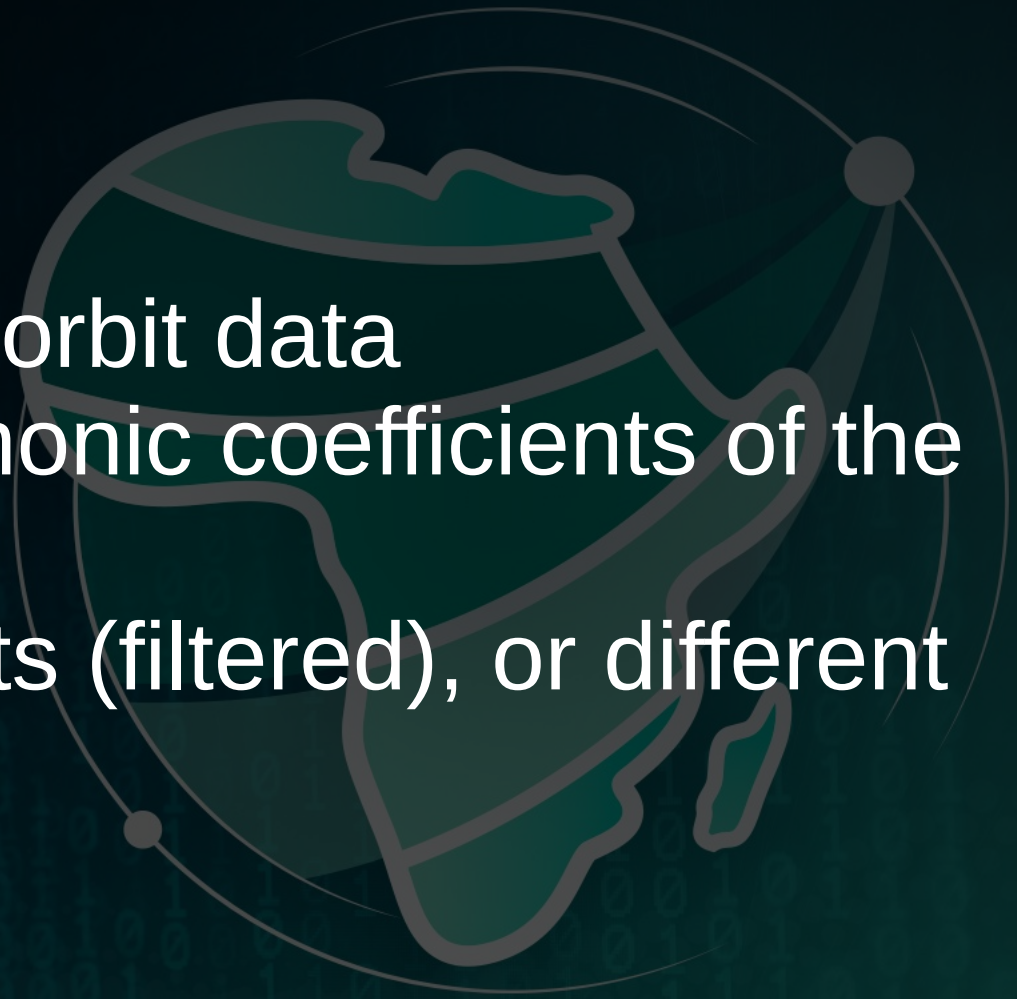


Working with satellite gravity data



Data products

- Level 1a/b: along track orbit data
- Level 2: Spherical harmonic coefficients of the gravity field
- Level3: gridded products (filtered), or different discretizations
-



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Working with monthly Level 2 GRACE/GRACE-FO products...

```
Unconstrained GRACE-only monthly gravity field solution computed at TU Graz

When using the data, please cite it as follows:
Mayer-GÃ¼rr, Torsten; Behzadpur, Saniya; Ellmer, Matthias; Kvas, Andreas;
Klinger, Beate; Strasser, Sebastian; Zehentner, Norbert (2018):
ITSG-Grace2018 - Monthly, Daily and Static Gravity Field Solutions from GRACE. GFZ Data Services.
http://doi.org/10.5880/ICGEM.2018.003

begin_of_head ===========================================================================

modelname               ITSG-Grace2018_n60_2003-07
product_type            gravity_field
earth_gravity_constant 3.9860044150e+14
radius                  6.3781363000e+06
max_degree              60
norm                    fully_normalized
tide_system             zero_tide
errors                  formal

key     L    M         C                   S                sigma C             sigma S
end_of_head ===========================================================================
gfc     0    0  1.000000000000e+00  0.000000000000e+00  0.000000000000e+00  0.000000000000e+00
gfc     1    0  0.000000000000e+00  0.000000000000e+00  0.000000000000e+00  0.000000000000e+00
gfc     1    1  0.000000000000e+00  0.000000000000e+00  0.000000000000e+00  0.000000000000e+00
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gfc     3    1  2.030468032807e-06  2.481957283426e-07  2.068553155579e-12  2.023927520327e-12
gfc     3    2  9.047786292834e-07 -6.190028755035e-07  2.158704256126e-12  2.056760542598e-12
```

Gravity products are provided as Stokes Coefficients

$$V(\theta,\lambda,r) = \frac{GM}{a}\sum_{n=0}^{\infty}\sum_{m=0}^{m=n}\left(\frac{a}{r}\right)^{n+1} P_{nm}(\cos\theta)\left(C_{nm}\cos m\lambda + S_{nm}\sin m\lambda\right)$$



- The actual information of the gravity field is contained in Cnm, Snm (unitless)
- The challenge with satellite gravimetry, strong attenuation of Potential with satellite altitude and degree:

$$\left(\frac{a}{r}\right)^{n+1}$$

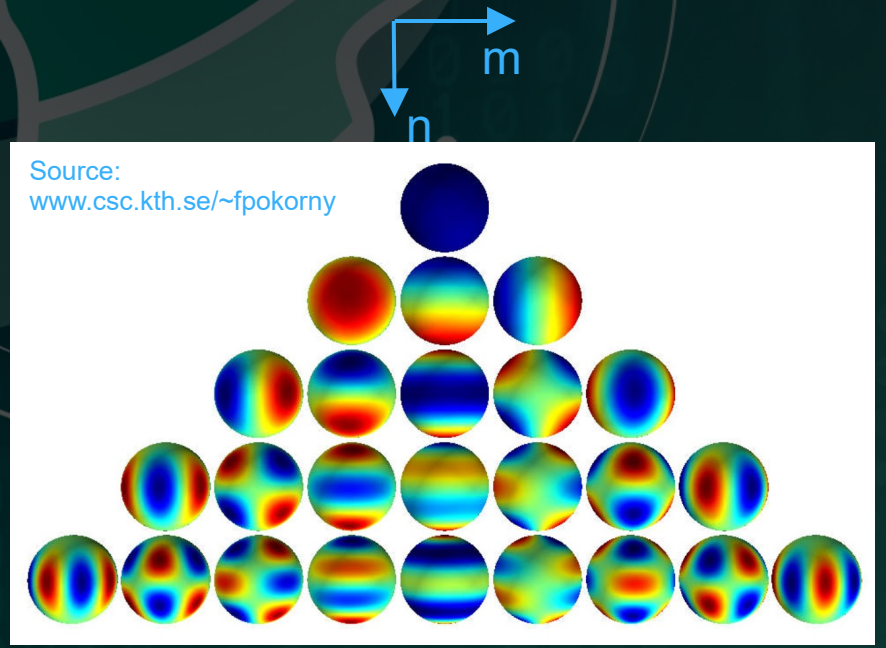
Gravity potential and spherical harmonics

$$V(\theta,\lambda,r) = \frac{GM}{a}\sum_{n=0}^{\infty}\sum_{m=-m}^{m=n}\left(\frac{a}{r}\right)^{n+1} C_{nm}Y_{nm}(\theta,\lambda)$$

$$Y_{nm}(\theta,\lambda) = \begin{cases} \bar{P}_{nm}(\cos\theta)\cos m\lambda), & m >= 0 \\ \bar{P}_{nm}(\cos\theta)\sin m\lambda), & m < 0 \end{cases}$$

- Spherical harmonics (bases function) are solutions of the Laplace equations!
- Higher degree (n) and order (m) allows resolving finer details of the gravity field

  Resolution ~40000km/(2*maxdegree)



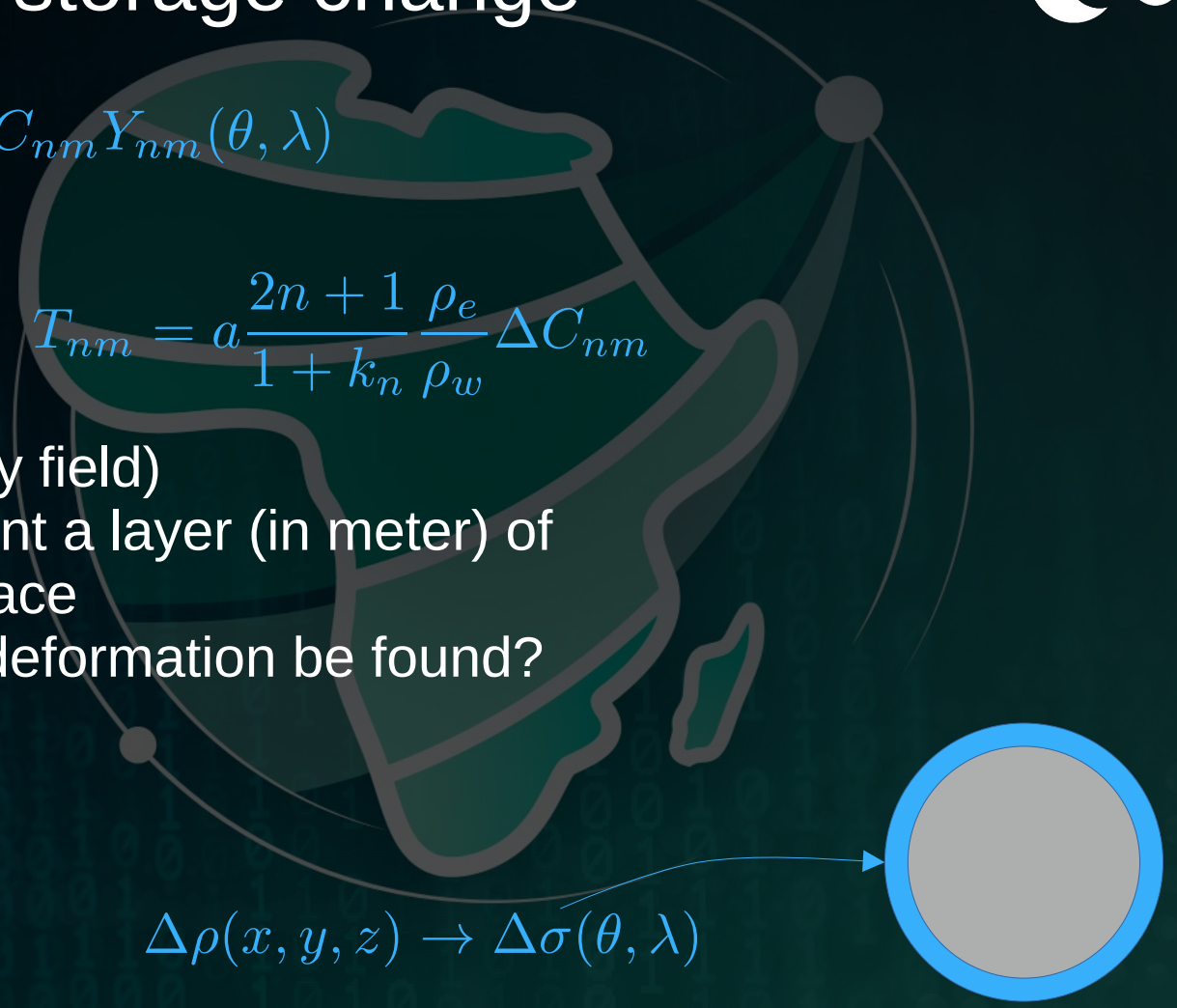
Conversion to terrestrial water storage change

$$TWS(\theta,\lambda) = a\sum_{n=0}^{\infty}\sum_{m=0}^{m=n}\frac{2n+1}{1+k_n}\frac{\rho_e}{\rho_w}\Delta C_{nm}Y_{nm}(\theta,\lambda)$$

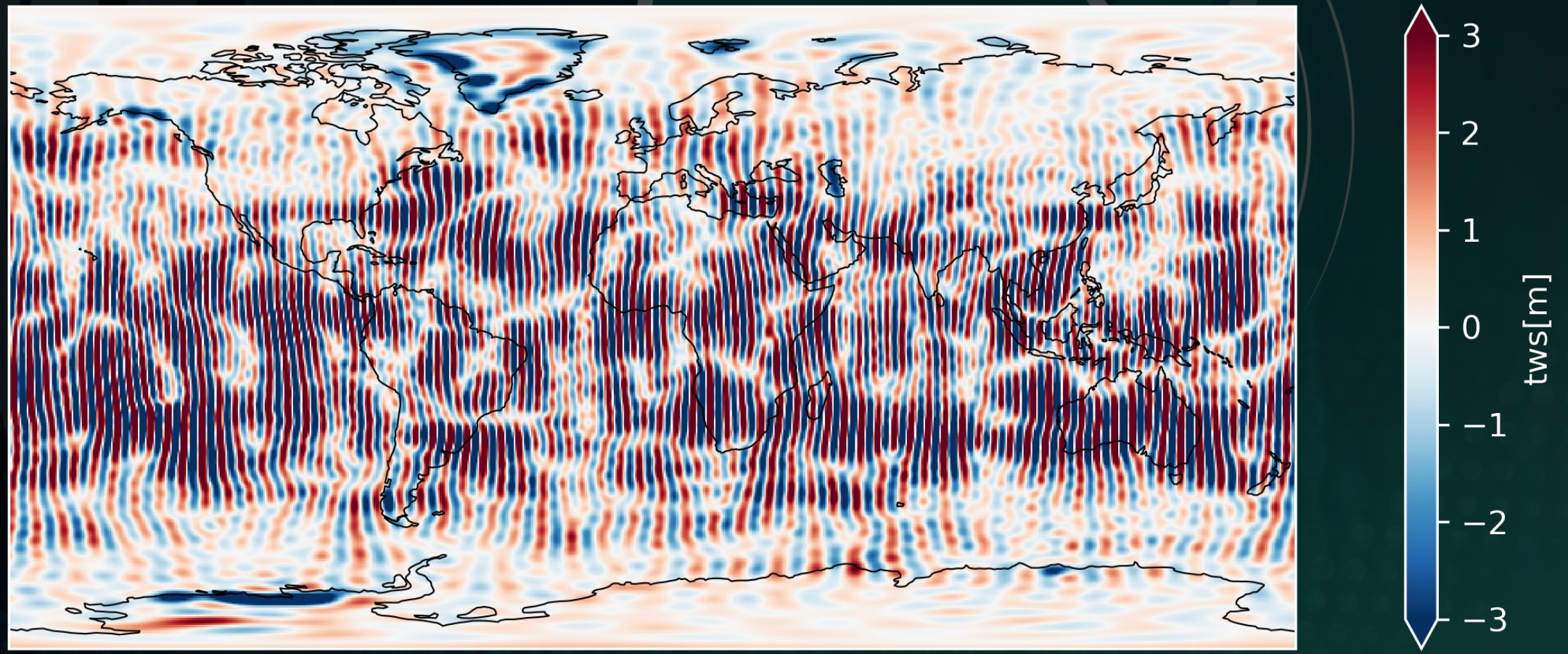
$$= \sum_{n=0}^{\infty}\sum_{m=0}^{m=n} T_{nm}Y_{nm}(\theta,\lambda), \qquad with \quad T_{nm} = a\frac{2n+1}{1+k_n}\frac{\rho_e}{\rho_w}\Delta C_{nm}$$

- Use anomalies (relative to a static gravity field)
- No dependency on radius, TWS represent a layer (in meter) of equivalent water height at the Earth's surface
- Where can the (indirect) effect of Earth deformation be found?

We assume that the mass variations of the gravity anomalies originate in an infinitely thin shell at the surface of the Earth (e.g. water,ice, air)



Stripes → Spatial/spectral filters are needed



W3, R. Rietbroek: TWS, OC10 Water budgets from space, EO AFRICA R&D Facility

Working with Level 2 GRACE/GRACE-FO products (tutorial)

- Other corrections needed (see tutorial)
  - Degree Coefficients 1 are missing
  - Coefficients with large n are too noisy and spatial filters are needed
  - Replace flattening coefficient (n=2,m=0) with that from Satellite Laser Ranging (better)
  - Static gravity field needs to be subtracted
  - For Ocean applications the a priori background model needs to be restored
- Many options available, but may be overwhelming for new users

Alternatively, Level 3/4 data (e.g. gridded, specific applications), Downside: no choice in spatial smoothing possible

- You can find level 3/4 products here:
  - Gravis service (Germany)
  - Tellus website (US)

shxarray

https://github.com/ITC-Water-Resources/shxarray

Main take-home points

- Terrestrial water storage (TWS) is a recognized Essential climate variable
- TWS is an aggregates water storage changes of different compartments of continental storage
- The time variable parts of TWS is detectable in the gravity field of the Earth
- The time-variable gravity field is very weak relative to the static anomalies
- Observed orbit anomalies provide information on the gravity field of the Earth
- GRACE and GRACE-FO carry inter-satellite ranging instruments which provide more information than just the orbits from GNSS tracking

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